Ex. 2. Two unbiased coins are tossed. What is the probability of getting at most one head ?

Sol. Here $S = \{HH, HT, TH, TT\}$.

Let E = event of getting at most one head.

$\therefore \quad E = \{TT, HT, TH\}$.

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}.$$

Ex. 3. An unbiased die is tossed. Find the probability of getting a multiple of 3.

Sol. Here $S = \{1, 2, 3, 4, 5, 6\}$.

Let E be the event of getting a multiple of 3.

Then, $E = \{3, 6\}$.

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

Ex. 4. In a simultaneous throw of a pair of dice, find the probability of getting a total more than 7.

Sol. Here, $n(S) = (6 \times 6) = 36$.

Let E = Event of getting a total more than 7

$= \{(2, 6), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6), (5, 3), (5, 4), (5, 5), (5, 6), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$.

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}.$$

Ex. 5. A bag contains 6 white and 4 black balls. Two balls are drawn at random. Find the probability that they are of the same colour.

Sol. Let S be the sample space. Then,

$n(s)$ = Number of ways of drawing 2 balls out of $(6 + 4) = {}^{10}C_2 = \frac{(10 \times 9)}{(2 \times 1)} = 45$.

Let E = Event of getting both balls of the same colour. Then,

$n(E)$ = Number of ways of drawing (2 balls out of 6) or (2 balls out of 4)

$$= ({}^{6}C_2 + {}^{4}C_2) = \frac{(6 \times 5)}{(2 \times 1)} + \frac{(4 \times 3)}{(2 \times 1)} = (15 + 6) = 21.$$

$$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{21}{45} = \frac{7}{15}.$$

Ex. 6. Two dice are thrown together. What is the probability that the sum of the numbers on the two faces is divisible by 4 or 6 ?

Sol. Clearly, $n(S) = 6 \times 6 = 36$.

Let E be the event that the sum of the numbers on the two faces is divisible by 4 or 6. Then

$E = \{(1, 3), (1, 5), (2, 2), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (5, 1), (5, 3), (6, 2), (6, 6)\}$

$\therefore \quad n(E) = 14$.

$$\text{Hence, } P(E) = \frac{n(E)}{n(S)} = \frac{14}{36} = \frac{7}{18}.$$

Ex. 7. Two cards are drawn at random from a pack of 52 cards. What is the probability that either both are black or both are queens ?

Sol. We have $n(s) = {}^{52}C_2 = \frac{(52 \times 51)}{(2 \times 1)} = 1326$.

Let A = event of getting both black cards;

B = event of getting both queens.

$\therefore \quad A \cap B$ = event of getting queens of black cards.

$$\therefore \quad n(A) = {}^{26}C_2 = \frac{(26 \times 25)}{(2 \times 1)} = 325, \; n(B) = {}^{4}C_2 = \frac{(4 \times 3)}{(2 \times 1)} = 6 \text{ and } n(A \cap B) = {}^{2}C_2 = 1.$$

$$\therefore \quad P(A) = \frac{n(A)}{n(S)} = \frac{325}{1326}; \; P(B) = \frac{n(B)}{n(S)} = \frac{6}{1326} \text{ and } P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{1326}.$$

$$\therefore \quad P(A \cup B) = P(A) + P(B) - P(A \cap B) = \left(\frac{325}{1326} + \frac{6}{1326} - \frac{1}{1326}\right) = \frac{330}{1326} = \frac{55}{221}.$$

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EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: *Mark (✓) against the correct answer:*

1. In a simultaneous throw of two coins, the probability of getting at least one head is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

2. Three unbiased coins are tossed. What is the probability of getting at least 2 heads?

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{1}{3}$ (d) $\frac{1}{8}$

3. Three unbiased coins are tossed. What is the probability of getting at most two heads?

(a) $\frac{3}{4}$ (b) $\frac{1}{4}$

(c) $\frac{3}{8}$ (d) $\frac{7}{8}$

4. In a single throw of a die, what is the probability of getting a number greater than 4 ?

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{2}{3}$ (d) $\frac{1}{4}$

5. In a simultaneous throw of two dice, what is the probability of getting a total of 7 ?

(a) $\frac{1}{6}$ (b) $\frac{1}{4}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

6. What is the probability of getting a sum 9 from two throws of a dice ?

(a) $\frac{1}{6}$ (b) $\frac{1}{8}$

(c) $\frac{1}{9}$ (d) $\frac{1}{12}$

7. In a simultaneous throw of two dice, what is the probability of getting a doublet?

(a) $\frac{1}{6}$ (b) $\frac{1}{4}$

(c) $\frac{2}{3}$ (d) $\frac{3}{7}$

8. In a simultaneous throw of two dice, what is the probability of getting a total of 10 or 11 ?

(a) $\frac{1}{4}$ (b) $\frac{1}{6}$

(c) $\frac{7}{12}$ (d) $\frac{5}{36}$

9. Two dice are thrown simultaneously. What is th probability of getting two numbers whose produc is even ?

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) $\frac{3}{8}$ (d) $\frac{5}{16}$

10. Tickets numbered 1 to 20 are mixed up and then ticket is drawn at random. What is the probabilit that the ticket drawn bears a number which is multiple of 3 ?

(a) $\frac{3}{10}$ (b) $\frac{3}{20}$

(c) $\frac{2}{5}$ (d) $\frac{1}{2}$

11. Tickets numbered 1 to 20 are mixed up and then ticket is drawn at random. What is the probabili that the ticket drawn has a number which is multiple of 3 or 5?

(a) $\frac{1}{2}$ (b) $\frac{2}{5}$

(c) $\frac{8}{15}$ (d) $\frac{9}{20}$

12. In a lottery, there are 10 prizes and 25 blanks. lottery is drawn at random. What is the probabili of getting a prize ?

(a) $\frac{1}{10}$ (b) $\frac{2}{5}$

(c) $\frac{2}{7}$ (d) $\frac{5}{7}$

13. One card is drawn at random from a pack of 5 cards. What is the probability that the card draw is a face card ?

(a) $\frac{1}{13}$ (b) $\frac{4}{13}$

(c) $\frac{1}{4}$ (d) $\frac{9}{52}$

14. A card is drawn from a pack of 52 cards. Th probability of getting a queen of club or a king heart is

(a) $\frac{1}{13}$ (b) $\frac{2}{13}$

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SOLUTIONS

1. Here S = {HH, HT, TH, TT}.

Let E = event of getting at least one head = {HT, TH, HH}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}$.

2. Here S = {TTT, TTH, THT, HTT, THH, HTH, HHT, HHH}.

Let E = event of getting at least two heads = {THH, HTH, HHT, HHH}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{4}{8} = \frac{1}{2}$.

3. Here S = {TTT, TTH, THT, HTT, THH, HTH, HHT, HHH}.

Let E = event of getting at most two heads.

Then, E = {TTT, TTH, THT, HTT, THH, HTH, HHT}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{7}{8}$.

4. When a die is thrown, we have S = {1, 2, 3, 4, 5, 6}.

Let E = event of getting a number greater than 4 = {5, 6}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}$.

5. We know that in a simultaneous throw of two dice, n (S) = 6 × 6 = 36.

Let E = event of getting a total of 7

= {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$.

6. In two throws of a die, n (S) = (6 × 6) = 36.

Let E = event of getting a sum 9

= {(3, 6), (4, 5), (5, 4), (6, 3)}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$.

7. In a simultaneous throw of two dice, n (S) = (6 × 6) = 36.

Let E = event of getting a doublet

= {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$.

8. In a simultaneous throw of two dice, we have n (S) = (6 × 6) = 36.

Let E = event of getting a total of 10 or 11

= {(4, 6), (5, 5), (6, 4), (5, 6), (6, 5)}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{5}{36}$.

9. In a simultaneous throw of two dice, we have n (S) = (6 × 6) = 36.

Let E = event of getting two numbers whose product is even.

Then, E = {(1, 2), (1, 4), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 4), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 4), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)}.

$\therefore \quad n(E) = 27$.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{27}{36} = \frac{3}{4}$.

10. Here, S = {1, 2, 3, 4, ..., 19, 20}.

Let E = event of getting a multiple of 3 = {3, 6, 9, 12, 15, 18}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{6}{20} = \frac{3}{10}$.

11. Here, S = {1, 2, 3, 4, ..., 19, 20}.

Let E = event of getting a multiple of 3 or 5 = {3, 6, 9, 12, 15, 18, 5, 10, 20}.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{9}{20}$.

12. P (getting a prize) $= \frac{10}{(10+25)} = \frac{10}{35} = \frac{2}{7}$.

13. Clearly, there are 52 cards, out of which there are 16 face cards.

$\therefore \quad P$ (getting a face card) $= \frac{16}{52} = \frac{4}{13}$.

14. Here, n (S) = 52.

Let E = event of getting a queen of club or a king of heart.

Then, $n(E) = 2$.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{2}{52} = \frac{1}{26}$.

15. Here, n (S) = 52.

There are 26 red cards (including 2 kings) and there are 2 more kings.

Let E = event of getting a red card or a king.

Then, $n(E) = 28$.

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{28}{52} = \frac{7}{13}$.

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